

VaaniAgent.

A voice agent that stays correct
when you change your mind.

4

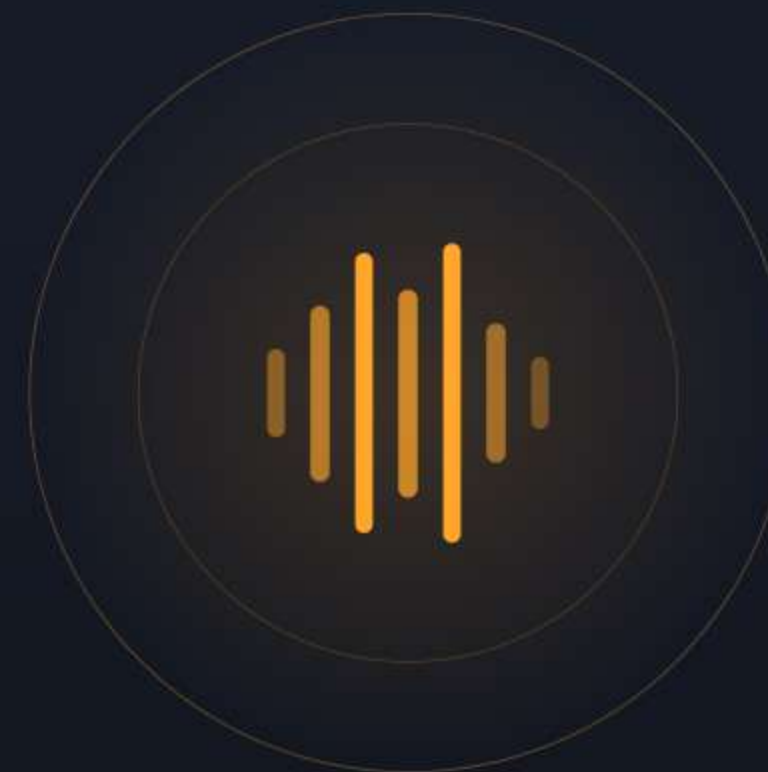
TEAM
MEMBERS

\$0

TOTAL API
COST

LIVE

DEPLOYED · VERCEL +
RENDER



TURN 1

“Find me hotels in Delhi under ₹5,000.”

TURN 2 · LATEST

“Actually, only vegetarian and near a metro.”

Only the latest request is spoken.

Voice assistants break when users change their mind.

LIVE CONVERSATION

USER

“Find me hotels in Delhi under ₹5,000.”

• • • Tool starts searching...

USER • MID-TOOL

“Actually, make it vegetarian and near a metro.”

The first search is still running – a ~4-5 s window.

THE PROBLEM

The old search may still finish — and speak a result the user no longer asked for.

OLD RESULT • turn 1

spoken anyway

Fixing this only at the audio level (“stop speaking”) leaves the state wrong. It has to be fixed at the state level.

01

USER INTERRUPTS

02

OLD TASK STILL RUNNING

03

STALE RESULT

04

WRONG RESPONSE

Hands busy. Eyes busy.

A professional in Delhi — commuting by metro or driving — searching hotels, restaurants and services without looking at a screen.



Hands-free and eyes-free is the whole point.

TEXT INTERFACE

Stop → Unlock the phone → Type the correction

VOICE

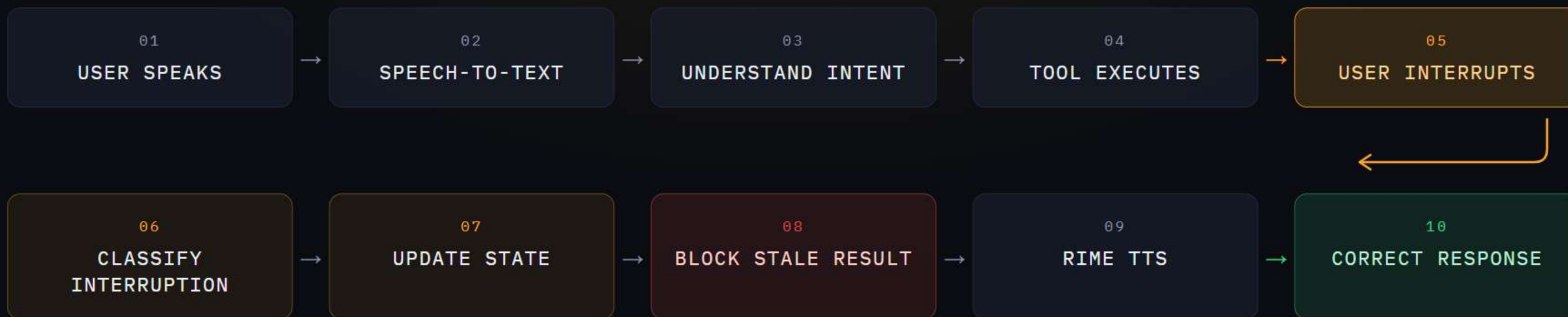
Speak naturally → Change the request mid-task →

Updated answer

Voice makes mid-task correction possible at all — removing voice would break the product.

Meet VaaniAgent

The interruption is not an error path — it is the main path, handled end to end. Live at vaani-agent.vercel.app.



THE CENTRAL IDEA

Only the latest valid task gets spoken.

Speech-to-text: Groq Whisper large-v3 (auto EN/HI) · Interrupt classify: `_classify_interrupt` shared with frontend `lib/interrupt.ts` · Voice: Rime only

Every interruption means something different.

Classified against the current task — three types fence the running tool, two never touch it.

CANCELS THE RUNNING TOOL

REFINE

“Actually, only vegetarian and near a metro”

Merge filters without losing context · re-run

CANCEL

“Forget it”

Fence in-flight tool · clear task state

PIVOT

“Find restaurants instead”

Switch hotel ↔ restaurant · invalidate old result

NEVER CANCELS THE TOOL — THE HARD PART

STATUS

TOOL KEEPS RUNNING

“What are you searching for?”

Answers progress while the search keeps running — async concurrency.

TOOL FUTURE		RUNNING
RIME STATUS		SPEAKING

The tool result still arrives, and is still spoken.

FACT Wikipedia aside

“Tell me about Connaught Place”

Speaks the summary · no cancel · no fence

The stale-result fence

```
if turn_id != state.turn_id: block
```

`turn_id` increments on every new user message. Results are fenced on `turn_id + active_request_id`.

TURN TIMELINE

TURN 1 · `turn_id = 1`
“Find hotels in Delhi”
`tool_status: RUNNING`



TURN 2 · `turn_id = 2`
“Make it vegetarian”
`turn_id++ · classify REFINE · tool_future`
`cancelled → CANCELLED`

`state.turn_id 2 · active_request_id req-2`
`tool_status RUNNING · stale_discarded 1`



STALE-
RESULT
FENCE

RESULT ARRIVES → CHECK `TURN_ID + REQUEST_ID`

OLD RESULT · `result.turn_id = 1` `1 ≠ 2`

✗ STALE RESULT BLOCKED

Silently discarded · `stale_discarded++` · never sent to Rime

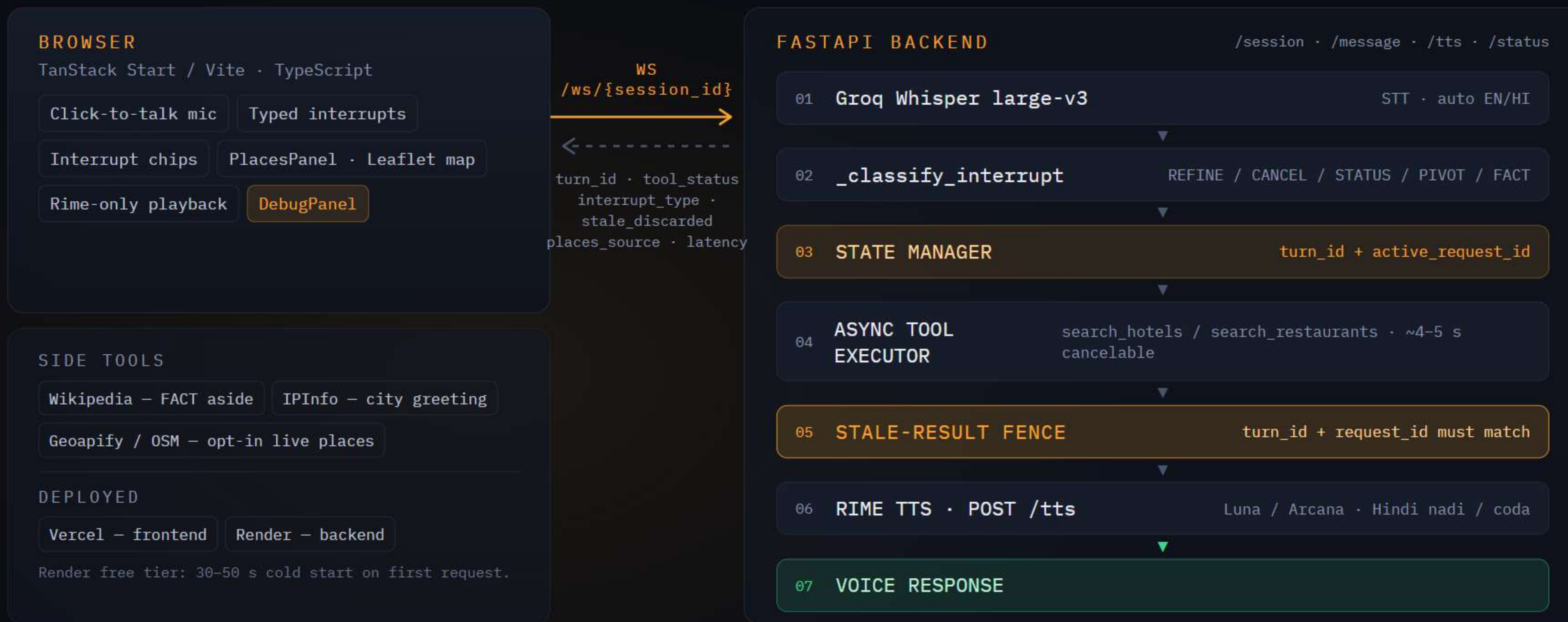
NEW RESULT · `result.turn_id = 2` `2 = 2`

✓ VALID → RIME SPEAKS IT

Only a matching `turn_id + request_id` reaches Rime TTS.

Old results never reach Rime.

How VaaniAgent works under the hood



Every spoken output is Rime – greeting, acknowledgements, status updates, FACT asides, results. No browser voice bridge.

Built around the hard part — not just voice.

01

5-Type Interrupt Classifier

REFINE CANCEL STATUS PIVOT FACT

One shared keyword source — backend `_classify_interrupt` mirrored by frontend `lib/interrupt.ts`.

02

India-First Context

IPInfo city greeting → Yes / Change city

Hinglish typed + spoken · auto EN/HI detection

₹ budget · veg · metro · cuisine · area filters

Separate Hindi Rime voice

03

Three-Layer Evidence

Metrics harness

pytest vs live ASGI app

Playwright Chromium smoke

Reproducible JSON any judge can re-run.

04

Live Debug Panel

Makes the invisible visible.

turn_id · tool_status

interrupt_type · stale_discarded

places_source · latency badges

05

Shipped, Not Slideware

`vaani-agent.vercel.app`

Deployed on Vercel + Render, running entirely on free tiers.

Honest by default: results are mock inventory so interrupts stay demoable; live OSM places are opt-in with no silent fallback.

We don't just claim it works. We test it.

THREE EVIDENCE LAYERS

01 · METRICS HARNESS

25 scripted scenarios across eight categories

`evaluation/results.json` · viewed at `/evaluate`

02 · PYTEST

Runs against the live ASGI app, mock places locked

`FACT` + deep mock edges → `qa_independent_results.json`

03 · CHROMIUM SMOKE

Playwright end-to-end on the real UI

`session` · `chips` · `typed search` · `STATUS` mid-flight · `map`

SCENARIO CATEGORIES

Normal

REFINE

CANCEL

STATUS

PIVOT

STALE

MULTI

EDGE

FIVE METRICS · TARGETS — RUN THE HARNESS FOR ACHIEVED NUMBERS

Interrupt Detection Accuracy

TARGET

85%+

Stale Response Block Rate

critical acceptance target · designed to block every stale result

TARGET

100%

Context Preservation Rate

TARGET

90%+

Interruption Recovery Latency

end of user speech → first Rime audio byte

TARGET

< 2000 ms

STATUS Non-Cancellation Rate

TARGET

100%

100% stale-result blocking is the critical acceptance target.

Scored suite is never fed a client-side `interrupt_type` — classification is the backend's own.

VaaniAgent — a voice agent that stays correct when you change your mind.

LIVE vaani-agent.vercel.app

Render free tier · allow 30-50 s cold start

JUDGE SCRIPT



Your latest request should always **win**.

